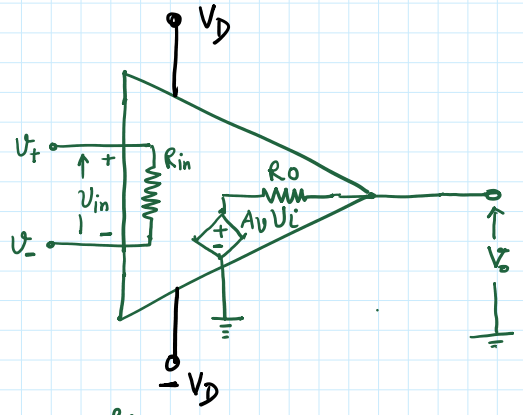
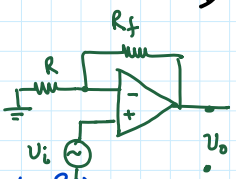
Lecture 33: OP-AMP Circuits and Applications

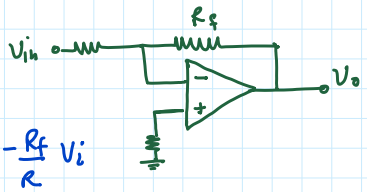
Recapitulation: OP-AMP is linear voltage amplifier.

$V_o = A_v V_i$





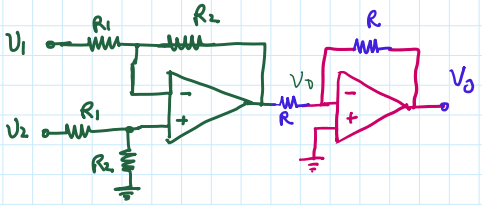
Various circuits - 1) Noninverting OP-Amp $V_o = \left(1+\frac{R_f}{R}\right) V_i$



2) Inverting OP-AMP $V_o = -\frac{R_f}{R} V_i$

3) Differential Amplifier $V_o = -(V_1 - V_2)$

$V_o = (V_1 - V_2)$



Common Mode Rejection Ratio: CMRR

A practical amplifier has output which is dependent on

1) voltage difference ; $d = V_1 - V_2$
2) Common mode voltage; $c = \frac{V_1+V_2}{2}$

A_d = Amplification for difference input

A_c = Amplification for common mode signal/input

Thus $V_o = A_d V_d + A_c V_c$

Then the ratio $\left|\frac{A_d}{A_c}\right| = \rho$ = CMRR = Common Mode Rejection Ratio

Thus, $CMRR = \left|\frac{A_d}{A_c}\right| = \rho$

$V_o = A_d V_d + A_c V_c = A_d V_d \left(1 + \frac{A_c}{A_d}\frac{V_c}{V_d}\right)$

$= A_d V_d \left(1 + \frac{1}{\left|\frac{A_d}{A_c}\right|}\frac{V_c}{V_d}\right)$

$$= A_d V_d \left(1 + \frac{1}{\left|\frac{A_d}{A_c}\right|} \frac{V_c}{V_d}\right)$$

$$V_o = A_d V_d \left(1 + \frac{1}{\rho} \cdot \frac{V_c}{V_d}\right)$$

CMRR can also be expressed in decibels as $20 \log_{10} \left|\frac{A_d}{A_c}\right|$

For example for $CMRR = 10^5$, $CMRR_{dB} = 20 \log_{10} 10^5 = 100\ dB$

High CMRR means $|A_d| >> |A_c|$ which boils down to design of transistors, completely identical in current gain conductance, threshold, base-emitter and temperature characteristics.

It needs high quality tail current, careful IC design, calibration, temperature compensation etc.

Increasing CMRR requires more transistors, more chip area, tighter fabrication band, complex packaging and greater power consumption.

High CMRR op-amp results in higher cost.

$CMRR = 1000$, $CMRR_{dB} = 20 \log_{10} 10^3 = 60\ db$ — low cost

$CMRR = 10{,}000$ $CMRR_{db} = 20 \log_{10} 10^4 = 80\ db$ ---- medium cost

$CMRR = 100{,}000$ $CMRR_{db} = 20 \log_{10} 10^5 = 100\ db$ - or higher ··High cost.

Question: OP-AMP

(A) $V_1 = 50\mu V$, $V_2 = -50\ \mu V \Rightarrow$ $V_d = V_1 - V_2 = 50 - (-50) = 100\ \mu V$

$V_c = \frac{V_1 + V_2}{2} = \frac{50 + (-50)}{2} = 0$

(B) $V_1 = 1050\mu V$, $V_2 = 950\mu V$

$\rho = 100, 1000, 10{,}000$

$V_o = ?$

$$V_o = A_d V_d \left(1 + \frac{1}{100} * \frac{0}{100 \times 10^{-6}}\right)$$

$$V_o = A_d \times 100 = 100\ A_d\ \mu V$$

(B) $V_1 = 1050\ \mu V$, $V_2 = 950\ \mu V$

$V_d = V_1 - V_2 = (1050 - 950) = 100\ \mu V$

$V_c = \frac{V_1 + V_2}{2} = \frac{1050 + 950}{2} = 1000\ \mu V$

$$V_o = A_d \times 100 \left(1 + \frac{1}{100} * \frac{1000}{100}\right)$$

$$V_o = A_d \times 100 \left(1 + \frac{1}{100} \times \frac{1000}{100}\right)$$

$$\boxed{V_o = 110\, A_d} = \boxed{V_o = A_v V_i}$$

$$V_o = A_d V_d + A_c V_c$$

$$V_o = A_v V_i + A_c V_c$$

$V_i = (V_+ - V_-) = (U_1 - U_2)$

$\beta = 1000$

$$V_o = A_d \times 100 \left(1 + \frac{1}{1000} \times \frac{1000}{100}\right)$$

$$= 101\, A_d \quad \mu V \qquad (1\%)$$

$$\beta = \left|\frac{A_d}{A_c}\right| = 10{,}000$$

$$V_o = A_d \times 100 \left(1 + \frac{1}{10000} \times \frac{1000}{100}\right) = A_d \times 100 \times 1.01$$

$$= 100.1\, A_d \quad \mu V$$

SOME OTHER OP-AMP CIRCUITS: Summing Circuits

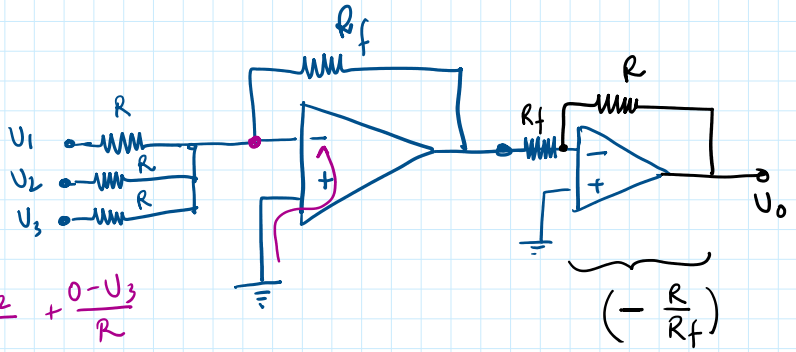
Integrator Circuit

Differentiator Circuit

Current to Voltage Converter

OP-AMP as active filters

Summing Circuit



$$\frac{0-U_1}{R} + \frac{0-U_2}{R} + \frac{0-U_3}{R} + \frac{0-V_o}{R_f} = 0$$

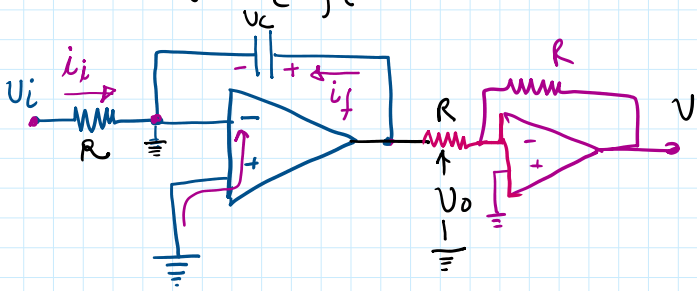
$$\Rightarrow \quad V_o = -\frac{R_f}{R}(U_1+U_2+U_3)$$

$$V_o' = -\frac{R_f}{R}\left(-\frac{R}{R_f}\right)(U_1+U_2+U_3)$$

$$\boxed{V_o' = U_1+U_2+U_3}$$

② Integrator : $V_o = \int V_i\,dt$

$V_c = \frac{1}{C}\int i\,dt \Rightarrow i_c = i_f = C\frac{dV_c}{dt} = C\frac{dV_o}{dt}$



KCL: $i_i + i_f = 0$

$\uparrow$

i_c

$$\frac{V_i}{R} + C\frac{dV_o}{dt} = 0$$

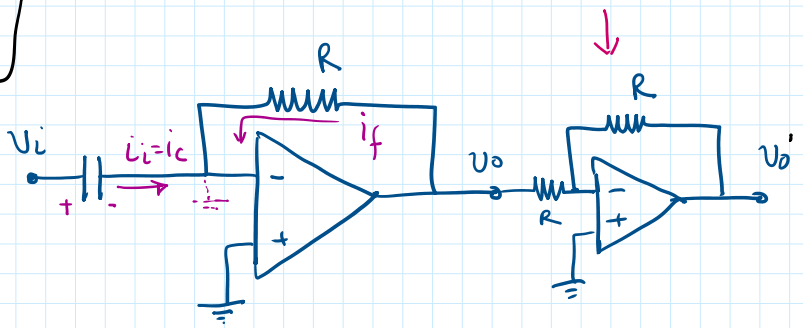
$$\Rightarrow \frac{dV_o}{dt} = -\frac{1}{RC}V_i$$

$$V_o = -\frac{1}{RC}\int V_i\,dt \quad ; \begin{matrix} R = 10\,k\Omega \\ C = 100\,\mu F \end{matrix} \Big\} \; RC = 1$$

$$= -\int V_i\,dt \quad \left(-\frac{R}{R}\right) = \int V_i\,dt$$

$$\boxed{V_o = \int V_i\,dt}$$

Differentiator :



$i_i + i_f = 0$

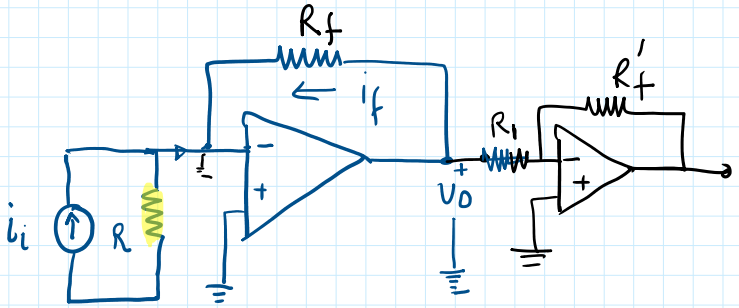
$\downarrow$

$$C\frac{dV_i}{dt} + \frac{V_o}{R} = 0$$

$$V_o = -RC\frac{dV_i}{dt} \quad ; \begin{matrix} R = 10\,k\Omega \\ C = 100\,\mu F \\ RC \simeq 1 \end{matrix}$$

$$\boxed{V_o = \frac{dV_i}{dt}}$$

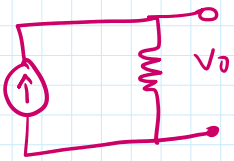
3. Current to voltage Convertor



$$i_i + \frac{V_o}{R_f} = 0$$

$$\Rightarrow V_o = \boxed{-R_f}\, i_i = \left(-R_f\right)\left(\frac{-R_f'}{R_1}\right) \Rightarrow \boxed{R_f R_f' = R_1}$$

④ Voltage follower $V_o = f(V_i) = \boxed{A_v} V_i$



$A_v = 1$

$V_o = V_i$